

Adham El-Messiry

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Professional Summary

Mechatronics Engineering student experienced in robotics, embedded systems, automation, and automotive diagnostics. Skilled at integrating software (Python, C++) with electromechanical hardware through sensor fusion, control logic, and system-level troubleshooting. Hands-on experience at AGMC BMW, AI robotics development, and mechanical restoration. Completed NVIDIA Generative AI and Cisco AI/IoT certifications to strengthen intelligent system design capabilities.

Education

Canadian University Dubai

Expected 2027

Bachelor of Science in Mechatronics Engineering

Focus: Robotics, Control Systems, Embedded Systems, Automation, Sensor Integration, Automotive Systems.

Certifications

NVIDIA – Generative AI Explained

Completed training on generative models, neural networks, and AI-driven system applications.

Cisco Networking Academy – Introduction to IoT

Studied IoT architecture, embedded systems, sensor integration, and cloud connectivity.

Cisco Networking Academy – Introduction to Modern AI

Apr 2026

- Learned core AI and Machine Learning concepts including object detection and image segmentation.
- Applied LLM concepts such as prompt engineering, summarization, and chatbot interaction.
- Explored real-world AI applications in automation, language models, and multimodal systems.

AI Autonomous Robocar (Applied Project Certification)

- Built a Raspberry Pi-based autonomous vehicle integrating camera and ultrasonic sensors.
- Implemented real-time navigation and lane detection using Python.
- Designed multi-sensor GPIO architecture with fast response time.

Experience

AGMC BMW — Engineering Intern

Sep 2024 – Nov 2024

Dubai, UAE

- Performed advanced diagnostics using BMW ISTA, including ECU fault tracing and CAN bus analysis.
- Assisted senior technicians with mechanical–electrical system repairs and recalibration.
- Coordinated workshop workflow, improving documentation accuracy and service efficiency.
- Conducted system testing and software-guided inspections.

- Developed brand identity, product pipeline, sourcing workflow, and quality control systems.
- Optimized digital marketing strategy and improved e-commerce performance.

Technical Projects

Smart Street Lighting System

- Designed LDR and motion sensor-based adaptive lighting system.
- Achieved 35% simulated energy reduction using embedded automation logic.
- Built scalable Raspberry Pi architecture with modular sensor integration.

Motorcycle Restoration and Electrical Rewiring

- Completed full mechanical overhaul and electrical harness reconstruction.
- Diagnosed ignition, charging, and starter systems using multimeter testing.
- Applied systematic fault isolation and repair methodologies.

Technical Skills

Programming: Python, C/C++, Embedded C, Arduino IDE

Hardware: Raspberry Pi, Microcontrollers, Sensors, Motor Drivers, ECU Systems, CAN Diagnostics

Tools: SolidWorks, MATLAB, AutoCAD, BMW ISTA, Multisim

Domains: Robotics, Automation, Mechatronics, Troubleshooting

Languages

Arabic (Native)

English (Fluent)

German (Basic)

Hobbies & Interests

Automotive restoration, robotics, AI systems, embedded development, sustainable technologies, and DIY mechatronics.